CAPSTONE PROJECT REPORT

FitToLend

AI Loan Eligibility Checker

An AI-Powered BFSI Financial Decision Platform

Submitted by: Kakre Karthik

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GitHub: github.com/karthik069069/BFSI-Project

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1. Abstract

FitToLend is an AI-powered BFSI (Banking, Financial Services and Insurance) web platform that simplifies personal financial decision-making. It brings four core tools into one seamless interface: a Loan Eligibility Checker, a Credit Score Analyzer, an EMI Calculator, and an AI Financial Tips module. The platform is built using HTML, CSS, and JavaScript with a dark glassmorphism design, and uses Claude AI to generate plain-language financial guidance from the numbers a user provides.

Users can check how much a lender is likely to offer them, understand what is helping or hurting their credit score, plan monthly repayments with live sliders, and ask follow-up questions to an AI assistant, all before approaching a bank. All calculations run in the browser, and saved calculations persisted in a private workspace when storage is available.

2. Introduction and Problem Statement

Applying for a loan is a high-stakes decision. Many applicants approach banks without knowing how much they can borrow, how their credit score will influence the offer, or what the monthly burden will be. A rejected application can itself leave a hard inquiry on the credit report and lower the score further.

Existing bank calculators are usually isolated: one for EMI, another for eligibility, with no explanation of why a result was produced. First-time borrowers in particular lack financial literacy to interpret raw numbers.

Problem statement: There is no simple, private and explainable tool that lets an individual check loan eligibility, estimate credit health, plan EMIs and receive personalized financial guidance in one place before applying for a loan.

3. Objectives

- Provide an instant, explainable loan eligibility estimate based on income, obligations, age, employment type, credit score, and existing loans.
- Help users understand their credit score, either by analysing a known score or by estimating one from credit behaviour.
- Offer a real-time EMI calculator with total interest and a year-wise amortization table.
- Generate personalised, plain-language financial tips and answer follow-up questions using AI.
- Deliver a modern, responsive, accessible interface that works on desktop and mobile.

4. Technology Stack

Layer	Technology / Purpose
Structure	HTML5, semantic sections, single-page layout with view switching
Styling	CSS3 with custom properties, glassmorphism (backdrop-filter), responsive grid and flexbox, reduced-motion support
Logic	Vanilla JavaScript (ES6+), no external framework
Typography	Google Fonts: Sora (headings), Inter (body), JetBrains Mono (figures)
AI	Claude AI integration for financial tips and follow-up Q&A, with a local rule-based fallback
Storage	Session-based secure storage (db) for saved calculations and recent activity
Hosting / VCS	GitHub (source code), Claude-hosted page for full AI features

5. System Overview and Architecture

FitToLend is a single-page application. A left navigation sidebar switches between five views (Overview and four modules) without reloading the page. A shared in-memory state object stores the latest result from each module so that the AI module can use it as context.

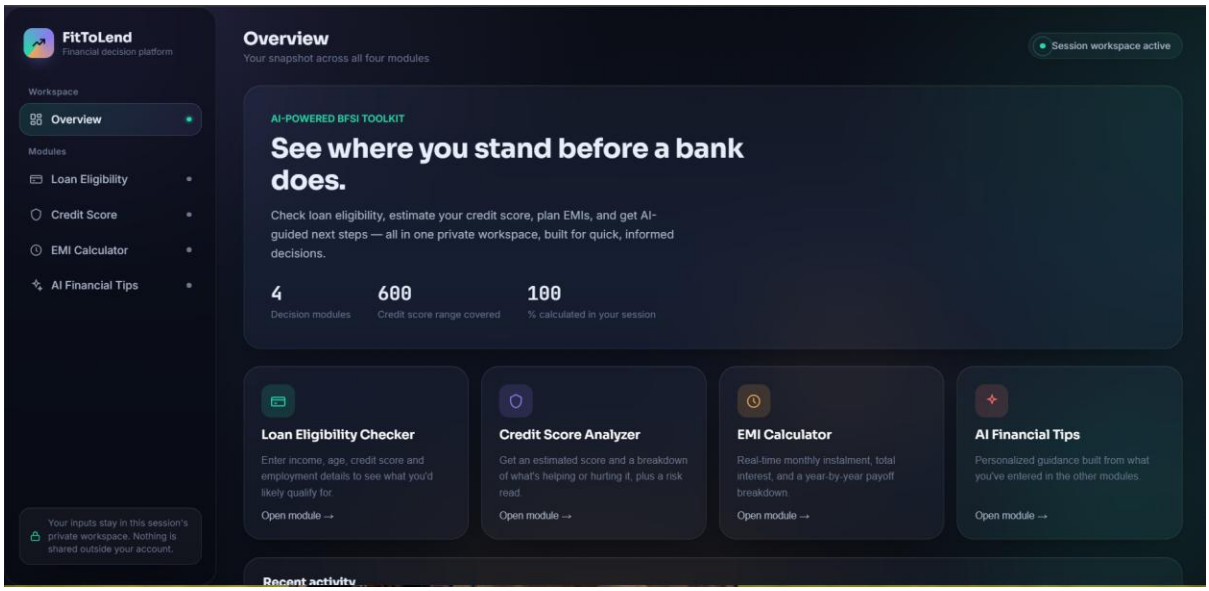
Component	Responsibility
Navigation shell	Sidebar, mobile drawer menu, page title and status indicator
Loan module	Validates inputs, computes eligibility, renders result with confidence meter and factor list
Credit module	Analyses a known score or estimates one from five behavioural factors
EMI module	Live two-way synced inputs and sliders, ring chart, KPI cards, amortization table
AI module	Builds a profile from user inputs and shared state, calls the AI, streams the answer
Shared state	Holds loan, credit and EMI results; feeds the AI profile and activity feed
Storage layer	Saves records to the workspace database; restores recent activity on load

Data flow: User input > client-side validation > calculation function > rendered result > shared state update > activity feed and storage > available as context for the AI module.

6. Module Descriptions

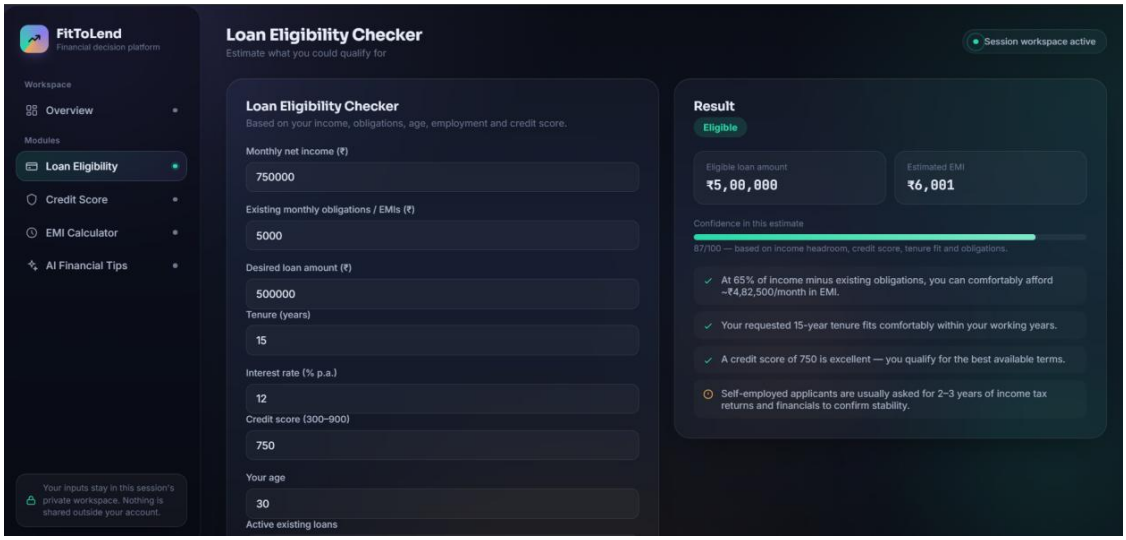
6.1 Dashboard / Overview

The landing page presents a hero section with animated statistics, four module cards that link to each tool, and a Recent Activity feed that lists the calculations performed in the session.



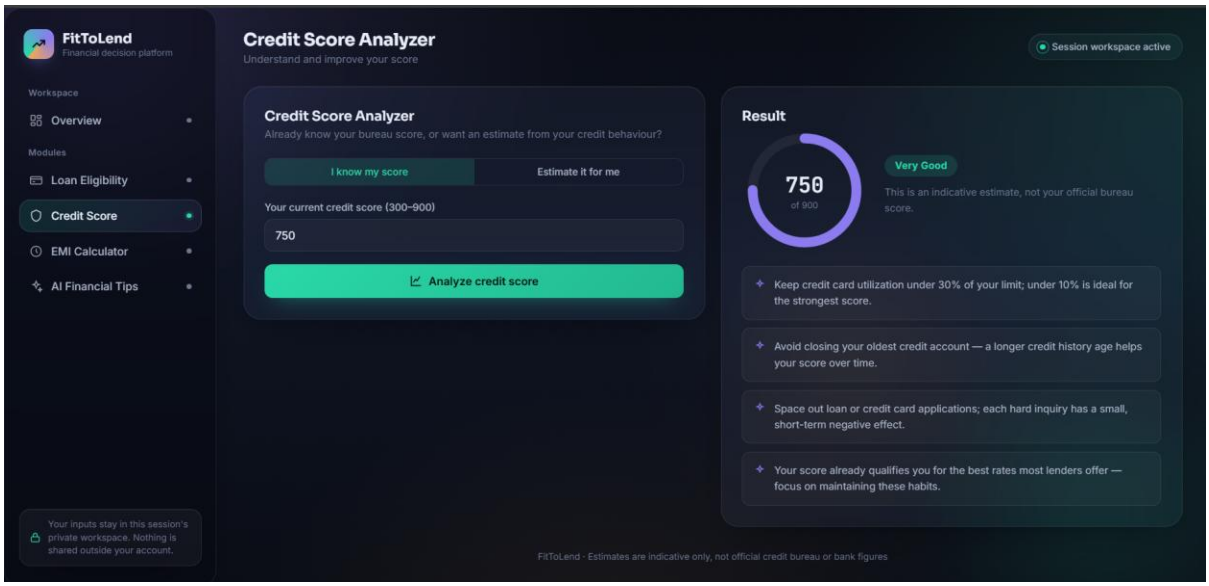
6.2 Loan Eligibility Checker

The user enters monthly net income, existing obligations, desired loan amount, tenure, interest rate, credit score (300 to 900), age, number of active loans and employment type. Every field is validated, with inline error messages. The result shows a status badge (Eligible, Conditionally Eligible or Not Eligible), the eligible amount, the estimated EMI, a 0 to 100 confidence meter and a list of factors, each marked positive, neutral or negative, explaining the outcome.



6.3 Credit Score Analyzer

Two modes are available. In "I know my score" mode, the score is categorized, and tailored improvement tips are shown. In "Estimate it for me" mode, the user supplies payment history, credit utilization, credit history age, number of open accounts and recent hard inquiries; the app estimates a score between 300 and 900 and displays a factor-by-factor breakdown. The result appears as an animated gauge with a category badge.



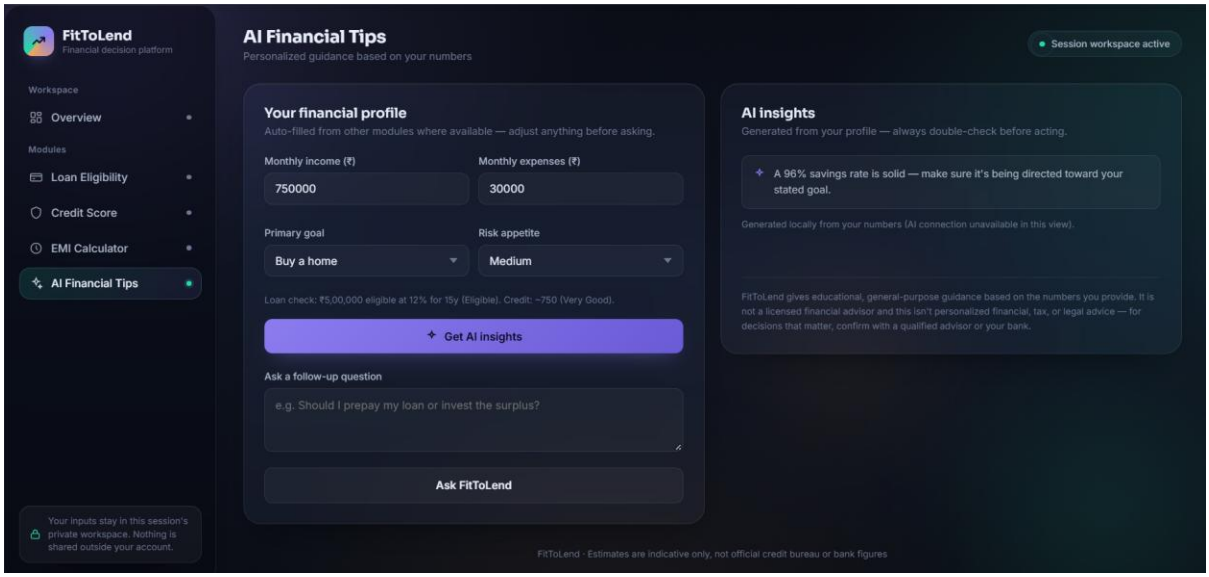
6.4 EMI Calculator

Loan amount, interest rate and tenure can be changed using number inputs or linked sliders, and results update instantly. Tenure can be switched between years and months. The module displays the monthly EMI, total interest, total payment, a principal-versus-interest ring chart, a guidance note and a year-wise amortization table. The user can save a calculation, which then becomes context for the AI module.



6.5 AI Financial Tips

The profile form is pre-filled from other modules where possible. "Get AI insights" returns four to six short, specific tips, and the follow-up box answers questions such as whether to prepay a loan or invest in the surplus. If the AI service is unavailable, a local rule-based generator produces tips from the same numbers, so the module never appears broken.



7. Algorithms and Formulas

7.1 EMI Formula

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EMI = P x r x (1 + r)^n / ((1 + r)^n - 1)
```

where P is the principal, r is the monthly interest rate (annual rate / 12 / 100) and n is the number of monthly instalments. For a 0% rate, EMI = P / n. Total payment = EMI x n and total interest = total payment - P.

7.2 Loan Eligibility Logic

- **FOIR band.** The Fixed Obligation to Income Ratio (FOIR) limit depends on income: 40% below Rs. 20,000; 45% from Rs. 20,000; 55% from Rs. 50,000; and 65% from Rs. 1,00,000 per month.
- **Affordable EMI.** Maximum affordable EMI = income x FOIR limit - existing obligations.
- **Age-based tenure cap.** The tenure is capped so that repayment ends by age 60 for salaried applicants and 65 for self-employed applicants.
- **Maximum principal.** The maximum principal is the present value of the affordable EMI over the effective tenure, which is the EMI formula solved for P.
- **Credit score multiplier.** Credit score below 650 multiplies the amount by 0.55; 650 to 699 by 0.80; 700 to 749 by 0.92; 750 and above by 1.00.
- **Existing loans.** Four or more active loans reduce the amount by a further 15%.
- **Employment gate.** Unemployed or student applicants are marked Not Eligible because there is no verifiable income.

- **Status rule.** Eligible if the offer is at least 90% of the request and the score is at least 650; Conditionally Eligible if at least 35% of the request; otherwise Not Eligible.
- **Confidence score.** A 3 to 98 score combining credit score (35 points), income headroom (30), tenure fit (15), employment type (up to 10) and loan count (10).

7.3 Credit Score Estimation

Each factor is scored from 0 to 100 and weighted. The weighted sum is then mapped onto the 300 to 900 range.

Factor	Weight	Scoring approach
Payment history	35%	On time = 100, 1-2 missed = 65, 3+ missed = 35, frequent defaults = 10
Credit utilisation	30%	At most 10% = 100, up to 30% = 85, 50% = 60, 75% = 35, above = 10
Credit history age	15%	7+ years = 100, 4+ = 80, 2+ = 60, 1+ = 40, otherwise 20
Account mix	10%	2 to 5 accounts = 100; 1 or 6 to 8 = 70; otherwise 40
Recent inquiries	10%	0 = 100, 1-2 = 80, 3-4 = 50, 5+ = 20

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Score = 300 + (weighted_total / 100) x 600
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Score categories: below 580 Poor; 580 to 669 Fair; 670 to 739 Good; 740 to 799 Very Good; 800 and above Excellent.

8. User Interface Design

- Dark glassmorphism theme with translucent panels, blur, soft gradients and subtle ambient animation.
- Colour semantics: teal for positive, amber for caution, red for negative, violet for AI and credit features.
- Responsive layout: the sidebar collapses into a slide-in drawer on screens narrower than 900 px, and two-column layouts stack on tablets and phones.
- Accessibility: visible keyboard focus outlines, labelled form fields, inline error messages and support for the reduced-motion preference.
- Typography pairs Sora for headings, Inter for body text and JetBrains Mono for numeric values.

9. AI Integration and Data Storage

9.1 AI integration

The AI module sends the user profile (income, expenses, goal, risk appetite and the saved loan, credit and EMI results) to Claude AI with an instruction to give short, concrete, cautious advice and to state that it is general guidance and not licensed financial advice. Responses stream into the interface as they are generated. If the AI connection is unavailable, a local fallback generator creates tips from the same data.

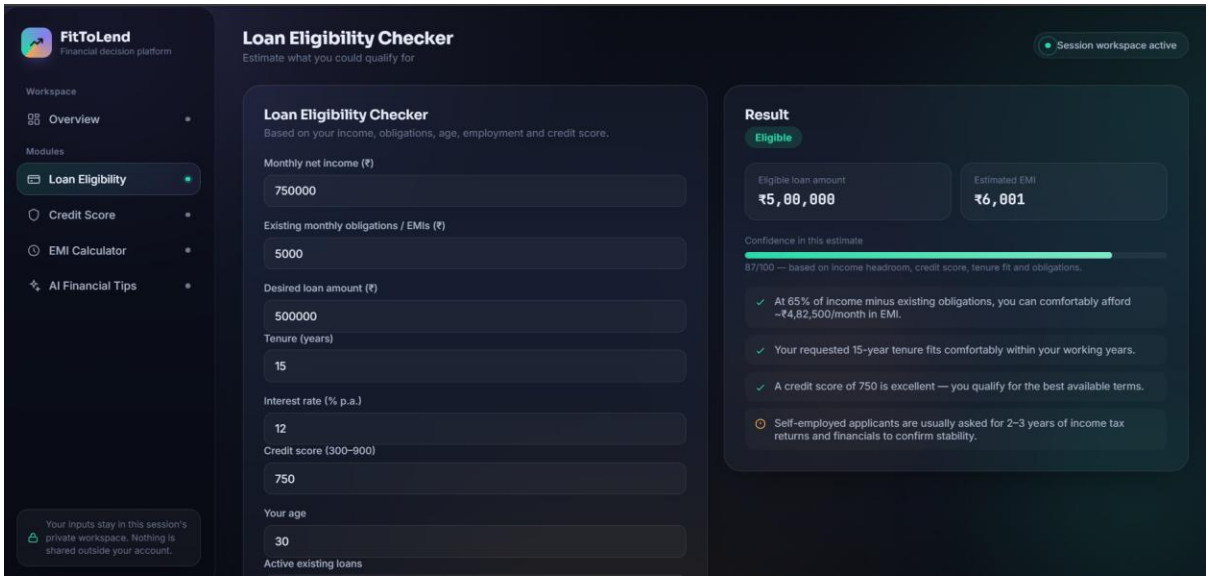
9.2 Data storage and privacy

Calculations run entirely in the browser. Saved loan, credit and EMI results are written to the user's private session workspace and the five most recent records are restored as activity when the page loads. Storage failures are handled silently so the main tools always keep working. No data is shared outside the user's account.

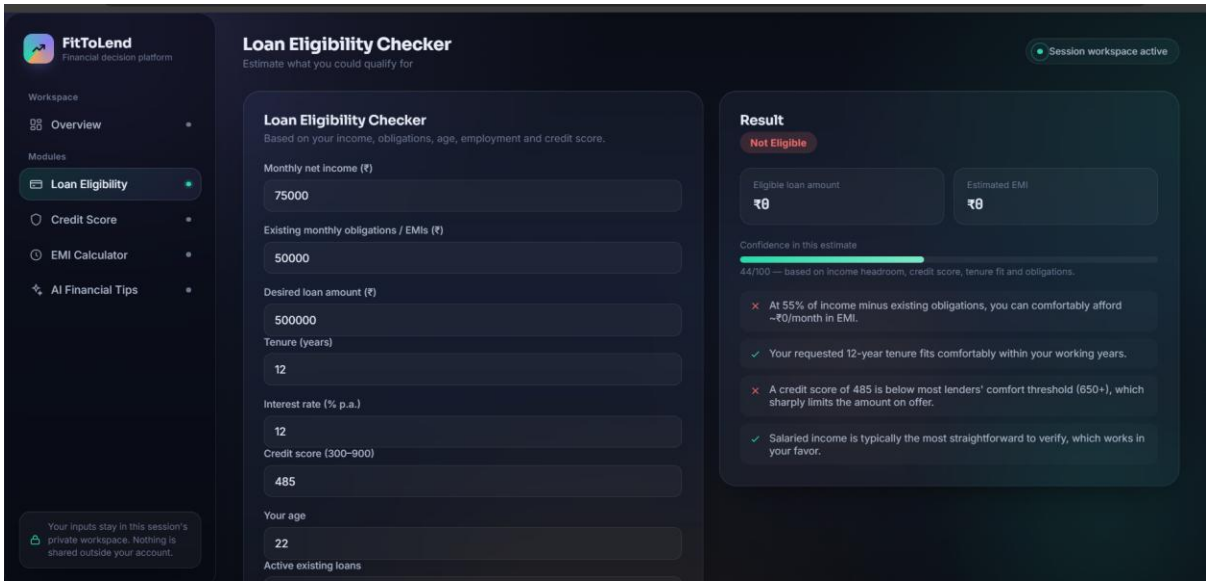
10. Testing and Sample Results

The following test cases were run through the same logic used by the application.

No.	Input	Expected behavior	Result
1	Private salaried; income Rs. 65,000; obligations Rs. 8,000; loan Rs. 25,00,000; 20 yrs; 10.5%; score 740; age 32; 1 loan	Full amount approved	Eligible; Rs. 25,00,000; EMI Rs. 24,959
2	Unemployed; income Rs. 40,000; loan Rs. 10,00,000	Rejected, no verifiable income	Not Eligible; Rs. 0
3	Private; income Rs. 30,000; obligations Rs. 5,000; loan Rs. 15,00,000; 15 yrs; 11%; score 600; age 45; 2 loans	Heavily reduced: low score and low-income headroom	Not Eligible; Rs. 4,11,316; EMI Rs. 4,675
4	EMI: Rs. 25,00,000; 9.5%; 20 years	EMI and interest computed	EMI Rs. 23,303; interest Rs. 30,92,787
5	Credit estimate: always on time; 30% utilisation; 4 yrs history; 3 accounts; 1 inquiry	High score	Estimated 843, Excellent
6	Credit: known score 782	Categorized correctly	Very Good
7	Loan form submitted with blank or out-of-range fields	Inline errors; no calculation	Fields highlighted red with messages
8	Open on a 375 px-wide screen	Sidebar becomes a drawer	Menu button appears; layouts stack



Test case: result (Eligible)



Test case: result (Not Eligible, reduced amount)

11. Limitations

- All outputs are indicative estimates. They are not official credit bureau scores or bank sanction decisions.
- Eligibility rules (FOIR bands, multipliers, age caps) are simplified models; real lenders use proprietary policies.

- AI responses and saved-record storage need the Claude-hosted environment. When the HTML file is opened locally, the AI module falls back to rule-based tips and records are not saved, while all calculators work fully.
- The credit estimator uses illustrative weights loosely inspired by common scoring factors, and is not calibrated to CIBIL, Experian or any other bureau.

12. Future Scope

- Export of saved calculations to Google Sheets or CSV through a small Apps Script or backend endpoint.
- A secure backend for AI calls, so the AI module works on any hosting platform.
- Loan comparison across banks, with live interest rate data.
- Prepayment and part-payment simulation in the EMI calculator, plus charts of balance over time.
- Insurance and investment (SIP) planning modules to extend the BFSI coverage.
- User accounts, PDF report download and multilingual support (Hindi, Telugu, Tamil and others).

13. Conclusion

FitToLend demonstrates how a clean interface, transparent financial logic and AI guidance can be combined into a practical decision-support tool for borrowers. Each result is accompanied by an explanation of the factors behind it, which improves financial literacy and helps users approach lenders better prepared. The project covers the BFSI domain end to end, from eligibility and credit health to repayment planning and personalized advice and provides a foundation for the extensions listed under Future Scope.

14. Submission Links

Item	Link
Source code (GitHub)	https://github.com/karthik069069/BFSI-Project
Demo video (Google Drive)	https://drive.google.com/drive/folders/1Ymyg40SbyL0dvm4BzX-C0h3prW8K1H6X?usp=sharing

Item	Link
Project report	FitToLend_Project_Report.docx / PDF in the GitHub repository

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